

CENTRE-OF-MASS EXTRACTION Using MATLAB

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Extracting the centre of mass of a binary pattern is an important task in pattern recognition area of digital image processing domain. The pattern's shape and size can be estimated using the radii profile that is computed around the centre of mass. The centre of mass of the binary pattern is computed using first-order moments of the pattern pixel's Cartesian coordinates. First, the pattern is converted into a binary class, i.e., pattern in black on a white colour background or vice

versa. In digital image processing, black is represented as '0' and white as '1' in binary form.

In MATLAB integrated development environment (IDE), image dimensions are expressed as number of rows x number of columns. The pixel at (0, 0) location is on the top left corner of the image (refer Fig. 1). An example pixel at location (3, 7) is shown as F(3, 7).

A grey-colour pattern of size 7x5 is shown in Fig. 2. It is converted into binary pattern as shown in Fig.

3. The grey pattern can be binarised using Otsu's algorithm. The equivalent binary pattern is shown in Fig. 3.

The binary image pattern is scanned from left to right, and top to bottom, i.e., in row x column manner. This is also called as 'raster scan.' This task is performed by using the `imread()` function in MATLAB. The `imread()` function enables binary image pattern in row x column matrix of '0' and '1' as black and white pixels, respectively. In Fig. 3, the matrix size is number of rows x number of columns = 15x15.

As stated before, the centre of mass of the pattern under test is computed using first-order moments of Cartesian coordinates of the pattern's pixels, i.e., black pixels or matrix elements with '0' value. First-order moments are computed using the following equations:

$$R_{CoM} = \frac{1}{N} \sum_{i=1}^N R_i \quad \text{and} \quad C_{CoM} = \frac{1}{N} \sum_{i=1}^N C_i$$

where (R_{CoM} , C_{CoM}) are row and column coordinates of the centre of mass of the pattern under test, (R_i , C_i) are coordinates of the pattern's i_{th} pixel, i.e., matrix elements with '0' value, and 'N' is the total number of pixels on the pattern body, i.e., the number of matrix elements with '0' value.

From Fig. 3, the centre of mass is as follows:

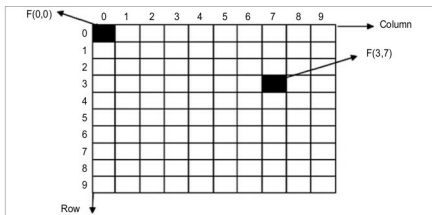


Fig. 1: Image as a matrix of rows and columns

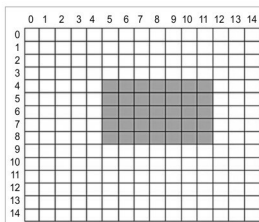


Fig. 2: Grey pattern

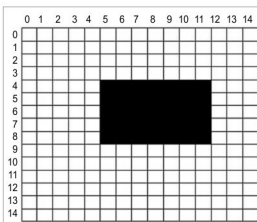


Fig. 3: Binary pattern